Sequences of Functions

Bartle & Sherbert §8.1 (8.1.1–8.1.10), §8.2 (8.2.1–8.2.2) · FYUGP pattern (April 2026), doubled

HOW THIS PAPER WORKS

A real FYUGP paper has 10 short (3 marks), 8 paragraph (6 marks) and 2 essay (10 marks, answer one) questions. This paper has **twice as many** (20 + 16 + 4), all from Module 4, and every concept of the module is the main point of at least one question.

Time	4 hours (or two sittings of 2 hours)
Maximum	140 marks (A ceiling 48, B ceiling 72, C answer any two)

PYQ MTS-25 Q9 = MTS 6B 10, March 2025, question 9 (old CBCSS course, same textbook). "also" = another paper asked the same thing. "adapted" = a 1-mark MCQ turned into a 3-mark question.

Section A · Short answer

All questions can be answered. Each carries 3 marks. Ceiling 48.

1. Define the uniform norm of a bounded function on a set A. Find $\|\sin x\|_{\mathbb{R}}$.

PYQ MTS-25 Q9

2. Define pointwise convergence of a sequence (f_n) of functions on a set A_0.

PYQ MTS-22 Q8 · also MAT-21 Q21

3. Define uniform convergence of a sequence of functions. How does it differ from pointwise convergence?

PYQ MTS-26 Q9

4. Evaluate $\lim e^{-nx}$ for $x \in \mathbb{R}$, $x \geq 0$.

PYQ MTS-25 Q10

5. Evaluate $\lim \dfrac{nx}{1+nx}$ for $x \in \mathbb{R}$, $x \geq 0$.

PYQ MTS-26 Q10

6. Evaluate $\lim\limits_{n\to\infty} \dfrac{x^2+nx}{n}$ for $x \in \mathbb{R}$.

PYQ MAT-22 Q8

7. Find $\lim \dfrac{\cos(nx+n)}{n}$, $x \in \mathbb{R}$.

PYQ MAT-21 Q10

8. Show that $F_n(x) = \dfrac{1}{n}\sin(nx+n)$ converges uniformly on $\mathbb{R}$ to 0, and find $\|F_n\|_{\mathbb{R}}$.

PYQ MTS-22 MCQ 12 · adapted

9. Discuss the uniform convergence of $f_n(x) = x^n$ on $(-1, 1]$.

PYQ MTS-22 Q9

10. If $h_n(x) = 2nx\,e^{-nx^2}$ for $x \in [0,1]$ and $h(x) = 0$, show that $\lim \int_0^1 h_n \neq \int_0^1 h$.

PYQ MTS-22 Q10

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11. Let $g_n(x) = x^n$, $x \in \mathbb{R}$. On which of the sets (−1, 1], (−2, 1], (0, 1], (−1, 1) does (g_n) converge to a continuous function? Justify.

PYQ MTS-22 MCQ 11 · adapted

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12. If (f_n) converges uniformly on $\mathbb{R}$ to f, must $(2f_n)$, (f_n^2) converge uniformly on $\mathbb{R}$? Justify.

PYQ MTS-22 MCQ 13 · adapted

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13. For $f_n(x) = x/n$ on $\mathbb{R}$, find a $K(\varepsilon, x)$ as in Lemma 8.1.3, and explain why it cannot be chosen independent of x.

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14. Show that $\dfrac{x}{x+n} \to 0$ for $x \geq 0$. Find $\|f_n\|$ on $[0, a]$ and on $[0, \infty)$.

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15. Show that $\lim \arctan(nx) = \dfrac{\pi}{2}\operatorname{sgn} x$ and deduce that the convergence is not uniform on $\mathbb{R}$.

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16. Does $\lim (\cos \pi x)^{2n}$ exist for all $x \in \mathbb{R}$? Find the limit function.

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17. Show that $x/n \to 0$ uniformly on [0, 1] but not on $\mathbb{R}$.

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18. Show that $x\,e^{-nx} \to 0$ uniformly on $[0, \infty)$.

19. Show that $x^n \to 0$ uniformly on $[0, \frac{1}{2}]$ and compute $\|x^n\|_{[0,1/2]}$.

20. State Theorem 8.2.2 (uniform limit of continuous functions) and show by an example that its converse is false.

Section B · Paragraph / problem

All questions can be answered. Each carries 6 marks. Ceiling 72.

21. Define pointwise convergence of a sequence of functions. Test the convergence of (x^n) on $\mathbb{R}$.

PYQ MAT-21 Q21

22. Discuss the convergence of $f_n(x) = x^n(1-x)$, $x \in [0,1]$.

PYQ MAT-21 Q32

23. Discuss the convergence of $f_n(x) = \dfrac{x^n}{x^n+1}$, $x \in [0,2]$.

PYQ MAT-22 Q31

24. Prove Lemma 8.1.5 (criterion for non-uniform convergence) and use it to show that $\dfrac{x^2+nx}{n}$ does not converge uniformly on $\mathbb{R}$.

25. For $g_n(x) = x^n$ on [0, 1], show that each g_n has a continuous derivative, that (g_n) converges pointwise, and that the limit is neither continuous nor differentiable at 1. Why does uniform convergence fail?

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26. Let $f_n(x) = \dfrac{nx}{1 + n^2x^2}$ on $[0, \infty)$. Show $f_n \to 0$ pointwise, find $\|f_n\|$, and decide whether the convergence is uniform on $[0, \infty)$ and on $[a, \infty)$, $a > 0$.

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27. Let f_n (n ≥ 2) be 0 on $[2/n, 1]$ and have a triangular graph with vertices (0, 0), (1/n, n), (2/n, 0) on [0, 2/n]. Show $f_n \to 0$ pointwise and $\int_0^1 f_n = 1$. What does this show?

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28. Show that $\dfrac{x^2 + nx}{n} \to x$ uniformly on [0, 8] and on every interval $[-b, b]$, but not on $\mathbb{R}$.

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29. Let $f_n(x) = x + \dfrac{1}{n}$. Show $f_n \to x$ uniformly on $\mathbb{R}$, that f_n^2 does not converge uniformly to x^2 on $\mathbb{R}$, but that it does on [0, 1].

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30. Find the pointwise limit of $f_n(x) = \dfrac{1}{1 + x^n}$ on [0, 1]. Is the convergence uniform? Find $\|f_n - f\|$.

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31. Using the Cauchy criterion, show that (x^n) does not converge uniformly on [0, 1], without finding its limit.

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32. Show that $F_n(x) = \frac{1}{n}\sin nx \to 0$ uniformly on $\mathbb{R}$, but that (F_n') does not converge. What does this show?

33. Show that $x^2 e^{-nx} \to 0$ uniformly on $[0, \infty)$ and find the norm.

34. Show that $n^2 x^2 e^{-nx} \to 0$ pointwise on $[0, \infty)$, not uniformly, but uniformly on $[a, \infty)$ for $a > 0$.

35. Let $f_n(x) = \frac{nx}{1+nx}$ on [0, 1]. Find the limit f, show the convergence is not uniform, and yet $\int_0^1 f_n \to \int_0^1 f$.

36. If $f_n \to f$ and $g_n \to g$ uniformly on A, prove $f_n + g_n \to f + g$ uniformly. If, in addition, all f_n, g_n are bounded, prove $f_n g_n \to fg$ uniformly.

Section C · Essay

Answer any two. Each carries 10 marks.

37. State and prove the Cauchy criterion for uniform convergence of a sequence of bounded functions.

PYQ MTS-25 Q26 · also MAT-21 Q38a

38. Let (f_n) be a sequence of continuous functions on $A \subseteq \mathbb{R}$ converging uniformly on A to $f : A \to \mathbb{R}$. Prove that f is continuous on A. Deduce that (x^n) is not uniformly convergent on [0, 1].

PYQ MTS-25 Q20

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39. Prove that a sequence (f_n) of bounded functions on $A \subseteq \mathbb{R}$ converges uniformly on A to f if and only if $\|f_n - f\|_A \to 0$. Apply it to x/n on [0, 1], and explain why it says nothing directly on $\mathbb{R}$.

PYQ MTS-26 Q20 **also MAT-22 Q19**

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40. Discuss the uniform convergence of (x^n) on [0, 1], on $[0, b]$ with $0 < b < 1$, and on [0, 1).

PYQ MAT-21 Q37b

Coverage

TOPIC	QUESTIONS
T1 Pointwise convergence	2, 13
T2 Finding the limit	4, 5, 6, 7, 16, 21
T3 Uniform convergence	3
T4 Proving not uniform	9, 24, 26, 34
T5 Uniform norm	1
T6 Norm test	39
T7 Computing the norm	14, 18, 22, 26, 33
T8 Bound free of x	8, 32
T9 Depends on the set	17, 19, 28, 40
T10 Cauchy criterion	31, 37
T11 Continuity lost	11, 25, 32
T12 Limits and integrals	10, 27, 35
T13 Theorem 8.2.2	38
T14 Using 8.2.2	15, 20, 23, 30
T15 Sums and products	12, 29, 36

All 21 in-syllabus past questions appear (16 directly, 2 as "also", 3 MCQs adapted).